

EIT-kit Demo: An Electrical Impedance Tomography Toolkit for Health and Motion Sensing

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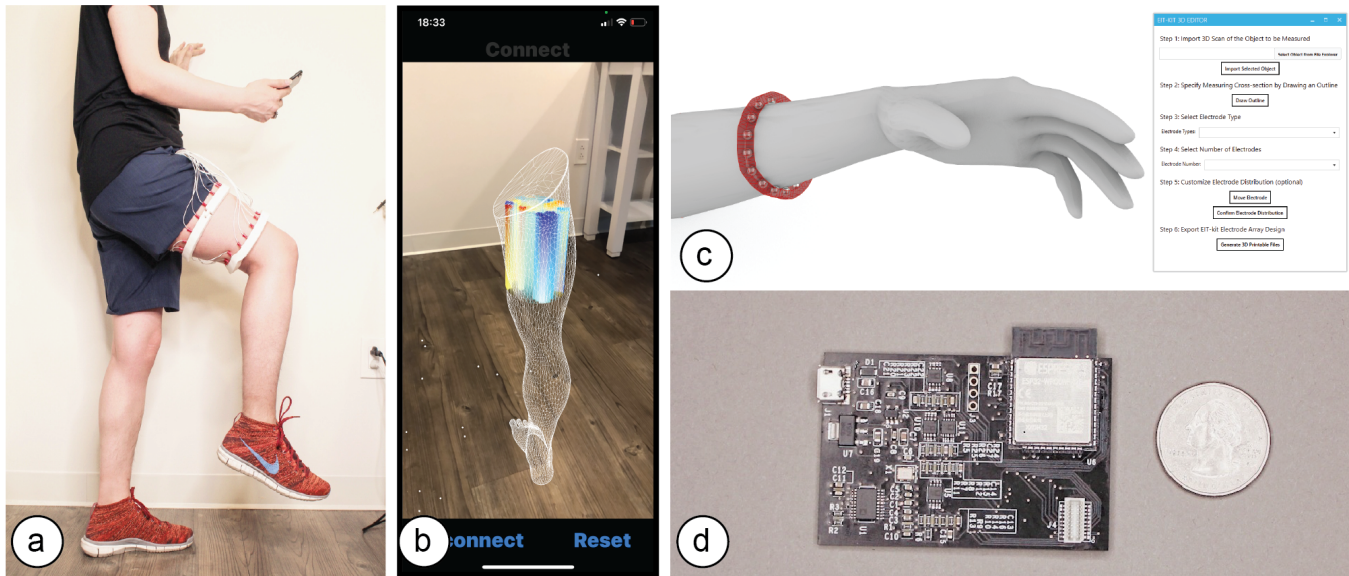


Figure 1: (a) EIT-kit supports users in creating a variety of EIT sensing devices and (b) visualizing the resulting data. EIT-kit provides (c) a 3D editor plugin, (d) a sensing motherboard and microcontroller library, as well as an image reconstruction API.

ABSTRACT

In this demo, we present *EIT-kit*, an electrical impedance tomography toolkit for designing and fabricating health and motion sensing devices. *EIT-kit* contains (1) an extension to a 3D editor for personalizing the form factor of the electrode arrays and the electrode distribution, (2) a customized EIT sensing motherboard that users can use to perform measurements, (3) a microcontroller library that automates electrical impedance measurements, and (4) an image reconstruction library for mobile devices for interpolating and then visualizing the measured data. Together, these allow for applications that require 2- or 4-terminal setups, up to 64-electrodes, and single or multiple (up to four) electrode arrays simultaneously.

We demonstrate each element of *EIT-kit*, as well as the design space that *EIT-kit* enables by showing various applications in health sensing as well as motion sensing and control.

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CCS CONCEPTS

• Human-centered computing → Interactive systems and tools.

KEYWORDS

electrical impedance tomography, health sensing, electronic prototyping, personal fabrication

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1 INTRODUCTION

Electrical Impedance Tomography (EIT) is an imaging technique that measures conductivity, permittivity, and impedance of a subject [5]. It works by attaching electrodes to the surface of the subject, and then using the electrodes to either inject current or measure the resulting voltages. Interpolating the raw signals then forms an image of the subject's internal conductivity.

In the past, EIT sensing required large, expensive hardware setups, as well as complicated image reconstruction algorithms. As a result, it was mainly used for professional medical applications in hospitals for monitoring the inner structure of a part of the human body [5], such as for observing lung function [4] and detecting cancer [14]. In the last decades, with the development of low-cost electronics and the availability of open-source EIT image reconstruction libraries, such as *EIDORS* [1], EIT sensing also became accessible to HCI researchers who have used it for touch sensing (*Electrick* [11]), tactile sensing [6], and hand gesture recognition (*Tomo* [10]). Such advances that made EIT sensing more portable have also fed back to the health sensing domain [2] where it has been used in sports medicine on the field [3] and home care [8], which demonstrates the great potential of low-cost EIT technology.

However, the expertise required for designing custom EIT devices is still high. To create an EIT device, users first have to design the form factor of the device to ensure constant contact between the electrodes and the subject. Next users have to connect the EIT device to a sensing board, which for custom applications they often have to design from scratch [12] [9]. On top of that users have to find the optimal AC injecting current and voltage measurement settings to optimize the signal acquisition. Finally, while there are visualization libraries for desktop applications (e.g. *EIDORS* [1] based on *MATLAB* [7]), they do not work on mobile devices and thus users have to write their own visualizations for portable applications.

In this demo, we present *EIT-kit*, an electrical impedance tomography toolkit that supports users across the different stages of EIT device development.

2 THE EIT-KIT

EIT-kit supports users across different stages of EIT device development, and facilitates creation of custom EIT sensing applications.

2.1 3D Editor for EIT Device Geometry and Electrode Distribution

When users create a new EIT device, *EIT-kit*'s 3D editor supports users in creating a device form factor that ensures sufficient contact with the subject (Figure 2) and allows them to specify the sensing resolution via custom electrode distributions. Once user are satisfied with the design, the 3D editor generates the fabrication files for building the physical EIT device.

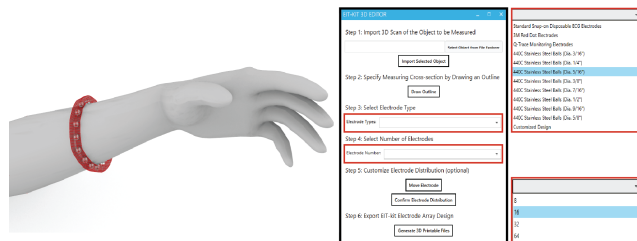


Figure 2: Device Form Factor: *EIT-kit*'s 3D editor generates the device geometry and electrode distribution automatically, here shown at the example of a wrist worn device on an imported arm 3D model.

2.2 EIT Sensing Motherboard

After fabricating the EIT device form factor, users next connect the EIT device to the *EIT-kit* sensing motherboard. The motherboard automates the EIT signal calibration and measurement for 2- and 4-terminal, with customizable number of electrodes (up to 64), and single or multiple electrode array (up to 4) configurations (Figure 3).

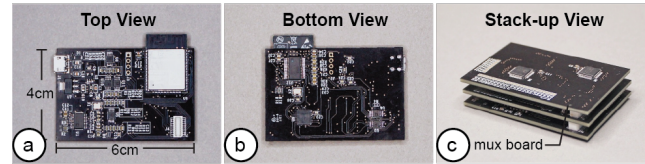


Figure 3: EIT sensing motherboard: (a) top view, (b) bottom view, (c) with two stacked up mux boards.

2.3 EIT Sensing Library

After connecting the EIT device to the sensing board, users next need to write a program to measure the signals. *EIT-kit* assists users by providing an EIT sensing library for the Arduino programming environment. Users first input the parameters of their measurement setup into the library's constructor and then call the library's `take_measurement()` function to start collecting data.

2.4 Image Reconstruction API

Once users started to collect data, they can use the functions from *EIT-kit*'s image reconstruction library to interpolate and then visualize the data (Figure 4). The image reconstruction library is built for mobile devices (e.g., iPhone, iPad) and can be used with the Xcode integrated development environment.

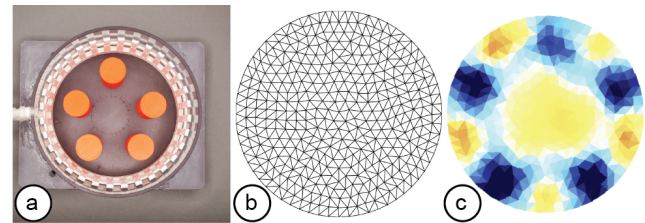


Figure 4: A reconstructed EIT image: (a) measurement setup, i.e. a water tank with five 3D printed PLA cylinders, (b) triangulated mesh, (c) reconstructed conductivity map generated on a mobile phone (model: iPhone XR).

3 CONCLUSION

We presented *EIT-kit*, an electrical impedance tomography toolkit for designing and fabricating health and motion sensing devices that supports users across the stages of EIT device development. We demonstrated the toolkit's applicability at the example of four interactive devices. For full content of the *EIT-kit* and implementation details, please refer to the full paper [13].

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